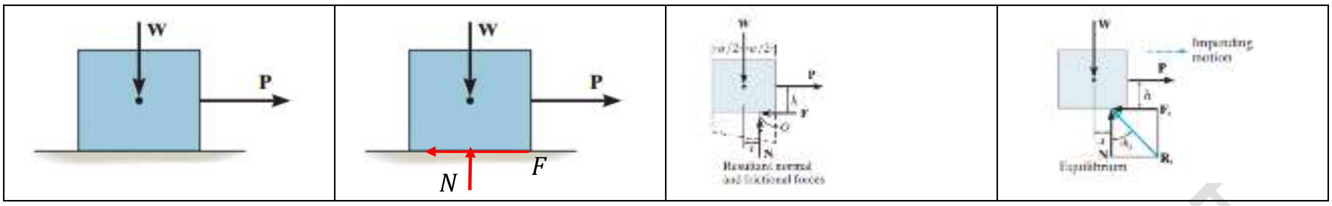
## Friction

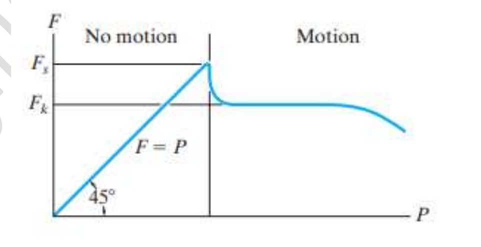
Friction is a force that resists the movement of two contacting surfaces that slide relative to one another. This force always acts tangent to the surface at the points of contact and is directed so as to oppose the impending or existing relative motion between the surfaces.




Consider a block of weight $W$ resting on a rough surface. Let a horizontal force $P$ acts on the block. If the value of $P$ is increased slowly from a low value, the block will not move initially. That is, the applied force $P$ is resisted by an equal force in the opposite direction. This resistive force is developed at the surface of contact which is known as the frictional force $F$. The force P and friction $F$ produces a moment in the clockwise direction. To maintain equilibrium, there should be an equal and opposite moment which will be developed by $W$ and $N$. Hence, the line of action of the normal force $N$ shifts to right as shown. The force $N$ can move only up to the edge of the block as we increase the force $P$. If the value of $P$ increases further, the body tips over.

From the diagram, $\tan \phi_s = \frac{F_s}{N}$. $\phi_s$ is known as angle of friction.

As the value of $P$ increases, the value of $F$ also increases and the body remains in equilibrium until it reaches a particular value, after which the body starts moving. Once the body starts moving, the frictional force decreases a bit. When the body is on the verge of motion, the value of friction is known as limiting static friction.




## Coulomb's laws of friction

1. Frictional force is independent of the area of contact provided the normal force is neither very small nor very high. When two surfaces are in contact, the surface irregularities will lock against each other which is resisting the relative motion. Though it appears that the entire surface is in contact, the actual contact area is only very small. The two surfaces can slide relative to each other only after breaking the locking surface irregularities. That is the reason why the frictional force decreases once the motion begins.

2. The frictional force is proportional to the normal force existing between the surfaces.

$$F \propto N \ggg F = \mu N$$

The constant $\mu$ is known as the coefficient of friction.

$$\mu = \tan \phi_s$$

3. For low velocity of sliding, the frictional force is independent of velocity.

### Example 1

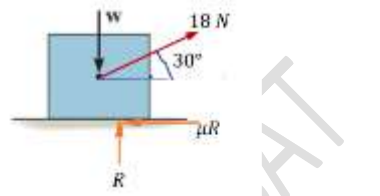
A body resting on a rough horizontal plane requires a pull of $18\ N$ inclined at $30°$ to the plane to just to move it. It was found that that a push of $22\ N$ inclined at $30°$ to the plane just moved the body. Determine the weight of the body and the coefficient of friction between the body and surface.

**Case 1**

$$\Sigma F_x = 0 \ \ggg\ 18\cos 30 - \mu R = 0 \ \ggg\ \mu R = 15.6$$

$$\Sigma F_y = 0 \ \ggg\ R + 18\sin 30 - W = 0 \gg R = W - 9$$

$$\mu(W-9) = 15.6 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

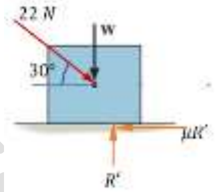



**Case 2**

$$\Sigma F_x = 0 \ \ggg\ 22\cos 30 - \mu R' = 0 \ \ggg\ \mu R' = 19.1$$

$$\Sigma F_y = 0 \ \ggg\ R'^{-22\sin 30 - W = 0} \gg R' = W + 11$$

$$\mu(W+11) = 19.1 \quad \ldots\ldots\ldots\ldots\ldots. (2)$$

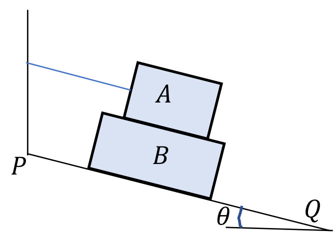



$$\frac{(1)}{(2)} \ \ggg\ \frac{\mu(W-9)}{\mu(W+11)} = \frac{15.6}{19.1} \ \ggg\ 19.1(W-9) = 15.6\,(W+11) \ \ggg\ W = \frac{15.6\times 11 + 19.1\times 9}{19.1 - 15.6} = 98.2\ N$$

$$(2) \ggg \quad \mu = \frac{19.1}{98.2+11} = 0.173$$

### Example 2

Find the value of angle that the surface $PQ$ makes with the horizontal, if block $B$ is about to slide down. The weight of the block $A$ is $100\ N$ and that of $B$ is $300\ N$. The block $A$ is tied to the wall with a string parallel to $PQ$. Take $\mu = 0.25$ for all contact surfaces. Also find the tension in the string.




Consider free body diagram of block $A$ and resolve the forces along the plane and perpendicular to it.

$$\Sigma F_x = 0 \ \ggg\ 100\sin\theta + \mu R - T = 0 \ \ggg\ T = 100\sin\theta + \mu R$$

$$\Sigma F_y = 0 \ \ggg\ R - 100\cos\theta = 0 \gg R = 100\cos\theta$$

$$T = 100\sin\theta + 0.25\times 100\cos\theta = 100\sin\theta + 25\cos\theta$$

Consider free body diagram of block $B$ and resolve the forces along the plane and perpendicular to it.

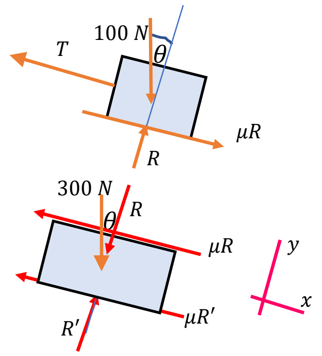
$$\Sigma F_x = 0 \ \ggg\ 300\sin\theta - \mu R - \mu R' = 0 \ \ggg\ \mu R' = 300\sin\theta - 0.25\times 100\cos\theta$$

$$\Sigma F_y = 0 \ \ggg\ R' - R - 300\cos\theta = 0 \gg R' = 300\cos\theta + 100\cos\theta = 400\cos\theta$$

$$0.25\times 400\cos\theta = 300\sin\theta - 0.25\times 100\cos\theta$$

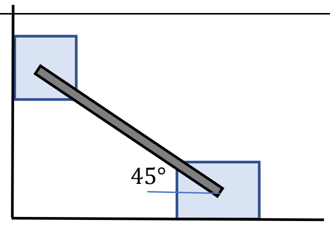
$$125\cos\theta = 300\sin\theta \ \ggg\ \tan\theta = 0.416 \ \ggg\ \theta = 22.6°$$

$$T = 100\sin\theta + 0.25\times 100\cos\theta = 61.5\text{N}$$



## Example 3

Two blocks $A$ and $B$ which are identical and weighing $100\ N$ each are supported by a rod at $45°$ as shown. If the blocks are in limiting equilibrium, find the coefficient of friction assuming it to be same for all surfaces.




**Equilibrium of block A**

$$\Sigma F_x = 0 \ggg R - F\cos 45 = 0 \ggg R = F\cos 45$$

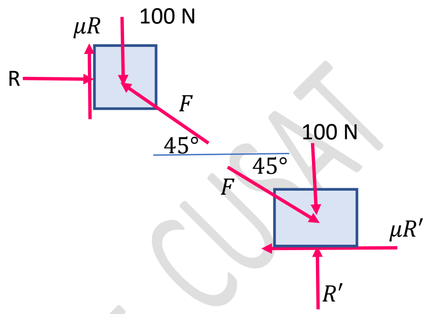
$$\Sigma F_y = 0 \ggg \mu R + F\sin 45 - 100 = 0 \ggg F(\mu + 1) = 100\sqrt{2}$$

**Equilibrium of block A**

$$\Sigma F_x = 0 \ggg F\cos 45 - \mu R' = 0 \ggg \mu R' = F\cos 45$$

$$\Sigma F_y = 0 \ggg R' - F\sin 45 - 100 = 0 \ggg R' = 100 + \frac{F}{\sqrt{2}}$$

$$\mu\left(100 + \frac{F}{\sqrt{2}}\right) = F\cos 45 \gg 100\mu = \frac{F}{\sqrt{2}} - \frac{F\mu}{\sqrt{2}} \gg F(1-\mu) = \mu 100\sqrt{2}$$




From the two equations

$$\frac{F(\mu+1)}{F(1-\mu)} = \frac{100\sqrt{2}}{\mu 100\sqrt{2}} \ggg \mu^2 + \mu = 1 - \mu \ggg \mu^2 + 2\mu - 1 = 0 \ggg \mu = 0.414$$

## Example 4

A uniform ladder of length $4m$ and weight $1800\ N$ rests on a horizontal floor and leans against a vertical wall at $45°$. The coefficient of friction between the wall and the ladder is $0.3$ and that between the ladder the floor is $0.5$. A man whose weight is $900\ N$ climbs up the ladder till it starts to slip. Determine to what height he can climb before it starts slipping?

Let the man climbs a distance $x$ before the ladder slips.

$$\Sigma F_x = 0 \ggg R - 0.5R' = 0 \ggg R = 0.5R'$$

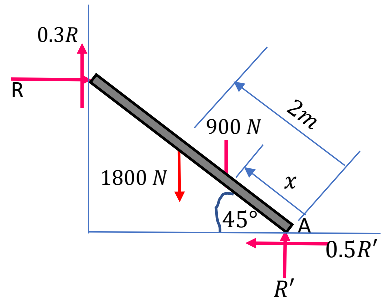
$$\Sigma F_y = 0 \ggg 0.3R + R' - 1800 - 900 = 0 \ggg 0.3 \times 0.5R' + R' = 2700$$

$$R' = \frac{2700}{0.15 + 1} = 2348\ N \quad \lll\ggg \quad R = 1174\ N$$

$\Sigma M_A = 0 \gg$

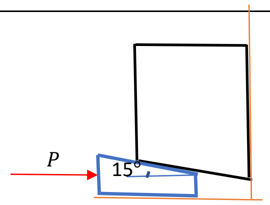
$$1800 \times 2\cos 45 + 900 \times x\cos 45 - R \times 4\sin 45 - 0.3R \times 4\cos 45 = 0$$

$$x = \frac{1174 \times 4\sin 45 + 0.3 \times 1174 \times 4\cos 45 - 3600\cos 45}{900\cos 45} = 2.78\ m$$




## Example 5

A wedge of angle $15°$ and negligible weight is used to raise a heavy block weighing $2000\ N$. Determine the minimum horizontal force $P$ necessary to just raise the block, if the coefficient of friction for all surfaces of contact is $0.25$.




**Equilibrium of the block**

$$\Sigma F_x = 0 \ggg R'\cos 75 + 0.25R'\cos 15 - R = 0 \ggg R = 0.5R'$$

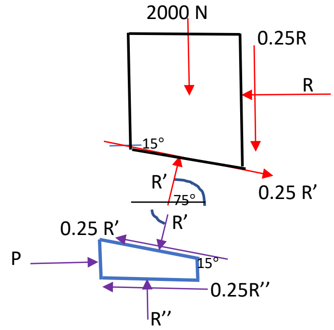
$$\Sigma F_y = 0 \gg R'\sin 75 - 0.25R'\sin 15 - 0.25R - 2000 = 0 \gg R' = 2576.6N$$

**Equilibrium of the wedge**

$$\Sigma F_x = 0 \ggg P - R'\cos 75 - 0.25R'\cos 15 - 0.25R'' = 0$$

$$\Sigma F_y = 0 \gg 0.25R'\sin 15 - R'\sin 75 + R'' = 0 \gg R'' = 2322.1\ N$$

$$P = 1870\ N$$



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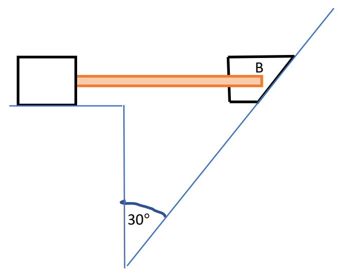
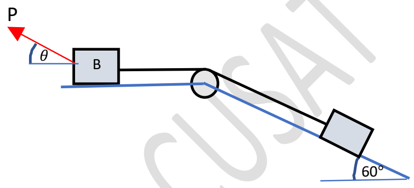
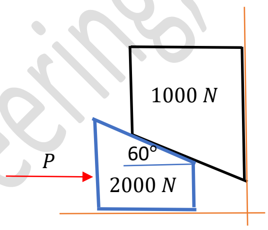
| Exercise 1<br>Two blocks which are connected by a horizontal link $AB$ are supported on rough planes as shown. The coefficient of friction between block $A$ and plane is $0.4$. The angle of friction for block $B$ on the inclined plane is $20°$. Find the smallest weight for the block A, for which equilibrium can exist, if the weight of block B is $500\ N$. |  |
| Exercise 2<br>Find the least force $P$ required to cause motion of the system shown to impend. Assume the coefficient of friction between all sliding surfaces as $0.2$, weight of block $A = 150\ N$ and weight of block $B = 100\ N$. Neglect the friction in the pulley. |  |
| Exercise 3<br>Referring to the figure, the coefficients of friction are as follows; $0.25$ at the floor, $0.3$ at the wall, and $0.2$ between the blocks. What is the minimum force that will hold the system in equilibrium? |  |

Reference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3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson